Lecture 23

Yang–Mills theory and instantons

Objectives. *Lecture 21 supplied the geometry of a gauge field. Lecture 22, in its section on higher Chern classes, gave* $c_2(F) = \operatorname{tr}(F \wedge F)/(8\pi^2)$ *for an* $\mathrm{SU}(r)$ *connection. We now choose an action and derive its field equation. For an* $\mathrm{SU}(2)$ *connection extending over the four-sphere, the curvature also gives an integer. This integer bounds the action and identifies solutions of a first-order equation. The BPST field fixes the orientation and normalization in a concrete example.*

By the end of this lecture you will be able to:

- Derive the vacuum Yang–Mills equation with the fundamental trace normalization.
- Prove the local Chern–Simons identity and distinguish it from global exactness.
- Complete the Euclidean action square and identify its equality condition.
- Compute the BPST action and charge and determine their behavior under orientation reversal.

Unit 23A The Yang–Mills field equation

23.1 The Yang–Mills action and equation

The passive frame change $e' = eh$ has $A^h = h^{-1}Ah + h^{-1}\mathrm{d}h$. We use this frame convention throughout. The derivative term distinguishes a potential from a form that transforms by conjugation alone.

Definition 23.1 (Adjoint-valued forms)**.** Lie-algebra-valued forms that transform by conjugation, with no derivative term, are called *adjoint-valued*.

We use a connection for $\mathrm{SU}(r)$, with $r \geq 2$, in its fundamental representation. We choose anti-Hermitian generators T^a normalized by $\operatorname{tr}(T^aT^b) = -\delta^{ab}/2$.[1] For $\mathrm{SU}(2)$, these are $T^a = -\mathrm{i}\sigma_a/2$ with the Pauli matrices of Lecture 21.[2] For spatial labels $a = 1$, $a = 2$ and $a = 3$, the indexed Pauli matrices are respectively σ_x, σ_y and σ_z. In Minkowski coordinates $x^0 = t$, the metric has signature $(+---)$ and the volume orientation is $\mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. The real component coefficients are defined by

$$A = T^a A^a_\mu \mathrm{d}x^\mu, \qquad F = \mathrm{d}A + A \wedge A = \frac{1}{2} T^a F^a_{\mu\nu} \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu.$$

[1] The Hermitian-potential dictionary is $A = -\mathrm{i}A_{\mathrm{Tong}}$ and $F = -\mathrm{i}F_{\mathrm{Tong}}$ for Tong's unrescaled matrices. His rescaling $A_{\mathrm{phys}} = A_{\mathrm{Tong}}/g_{\mathrm{YM}}$ instead gives $A = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$. Tong, §2.1, uses the positive Hermitian trace in his component action.

[2] The real basis from *Lie Groups and Their Representation Theory* is related to these generators by $(\mathrm{i}\sigma_z, \mathrm{i}\sigma_y, \mathrm{i}\sigma_x) = (-2T^3, -2T^2, -2T^1)$. That basis has cyclic bracket factor 2. The generators T^a used here have cyclic bracket factor one.

Here a runs over the $r^2 - 1$ generators. Spacetime indices run from 0 to 3. The coupling $g_{\mathrm{YM}} > 0$ is a scalar, distinct from the metric g. We write vol for the metric volume form denoted by μ in Lecture 15. Assume the displayed action integrals converge, or work on a bounded region with variations supported in its interior. The curvature transforms by conjugation under a frame change. Taking a trace of its Hodge pairing gives a gauge-invariant four-form. This quadratic expression generalizes the Maxwell action of Lecture 19. The term $A \wedge A$ in the curvature makes the nonabelian theory nonlinear.

Physical input. In classical Yang–Mills theory, the gauge field is a connection. Its vacuum dynamics are the stationary points of the action defined below, under compactly supported variations. For gauge group SU(3), this is the gauge-field part of the theory of the strong interaction. These physical inputs are described in Tong, §2.1–2.1.1.

Definition 23.2 (Yang–Mills action)**.** The *Yang–Mills action* is the following metric-dependent functional in Minkowski signature:

$$S_M[A] = \frac{1}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(F \wedge \star F) = -\frac{1}{4g_{\mathrm{YM}}^2} \int_M F^a_{\mu\nu} F^{a\mu\nu} \operatorname{vol}.$$

The covariant derivative on an endomorphism-valued k-form B is $DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A$. Here k denotes the form degree.

Theorem 23.3 (The field equation · *proved*)**.** *The Minkowski action S_M is gauge invariant. Its stationary points under arbitrary smooth compactly supported connection variations satisfy $D\star F = 0$.*

Proof. The Hodge pairing of two two-forms contracts their components with a factor $1/2$. Multiplying this by $\operatorname{tr}(T^a T^b) = -\delta^{ab}/2$ gives the coefficient $-1/4$. On overlaps and under a frame change, $F^h = h^{-1} F h$. The star acts only on differential forms, so conjugation and cyclicity of the trace make the integrand global and gauge invariant.

The difference of two connections transforms by conjugation, with no derivative term. Let $a = \delta A$ be a compactly supported adjoint-valued one-form. This a denotes a variation, whereas the superscript in T^a labels a generator. Differentiate $\mathrm{d}(A + \varepsilon a) + (A + \varepsilon a)^2$ at $\varepsilon = 0$. The result is $\delta F = \mathrm{d}a + A \wedge a + a \wedge A = Da$. Symmetry of the two-form pairing gives

$$\delta S_M = \frac{2}{g_{\mathrm{YM}}^2} \int_M \operatorname{tr}(Da \wedge \star F),$$
$$\operatorname{tr}(Da \wedge \star F) = \mathrm{d} \operatorname{tr}(a \wedge \star F) + \operatorname{tr}(a \wedge D\star F).$$

The second line follows from the graded product rule. The connection commutators cancel inside the trace. Stokes removes the compactly supported exact term. The remaining pairing with every a vanishes exactly when $D\star F = 0$. The trace pairing is nondegenerate on $\mathfrak{su}(r)$, so it detects every component. ∎

With matter, define the connection-valued current three-form $\mathcal{J}_{\mathrm{conn}}$ by $\delta S_{\mathrm{matter}} = -(2/g_{\mathrm{YM}}^2) \int_M \operatorname{tr}(a \wedge \mathcal{J}_{\mathrm{conn}})$. The same variation then gives $D\star F = \mathcal{J}_{\mathrm{conn}}$.

For U(1) the generator is $-\mathrm{i}$, whose trace square is -1. We use the different prefactor $1/(2g^2)$, where $g = q/\hbar \neq 0$ is signed. This abelian coefficient differs from the positive nonabelian coupling g_{YM}. Lecture 21's physical-potential convention gives

$$A = -\frac{\mathrm{i}q}{\hbar} A_{\mathrm{em}}, \qquad F = -\frac{\mathrm{i}q}{\hbar} F_{\mathrm{em}}, \qquad S_M = \frac{1}{2g^2} \int_M F \wedge \star F = -\frac{1}{2} \int_M F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}.$$

Let $\mathcal{J} = \star j$ be Lecture 15's real current and set $\mathcal{J}_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)\mathcal{J}$. For this prefactor, with $a = \delta A$, define the matter current by

$$\delta S_{\mathrm{matter}} = -\frac{1}{g^2}\int_M a \wedge \mathcal{J}_{\mathrm{conn}} = \int_M \delta A_{\mathrm{em}} \wedge \mathcal{J}.$$

The equation $\mathrm{d}\star F = \mathcal{J}_{\mathrm{conn}}$ becomes Lecture 15's $\mathrm{d}\star F_{\mathrm{em}} = \mathcal{J}$. The real current and the imaginary connection current are distinct.

Self-check. Show directly that $\mathrm{tr}(F \wedge \star F)$ is invariant under a frame change. Explain why this does not require the two local potentials to be equal.

Answer. The transformed product is $h^{-1}(F \wedge \star F)h$, since the star commutes with matrix conjugation. Its trace equals the original trace by cyclicity. The potentials obey the affine frame law, which also contains $h^{-1}\mathrm{d}h$. Gauge invariance concerns the traced curvature expression, rather than equality of potentials.

Unit 23B Euclidean duality, Chern–Simons and the bound

23.2 Euclidean action and duality

We now study a Riemannian variational problem on an oriented four-manifold. Its positive action compares fields with the same second Chern number. We replace the Minkowski metric by a positive Riemannian metric.

Definition 23.4 (Euclidean action). The *Euclidean action* is

$$S_E[A] = -\frac{1}{g_{\mathrm{YM}}^2}\int_M \mathrm{tr}(F \wedge \star F).$$

The minus sign makes the density nonnegative. In an orthonormal form basis, the matrix pairing is $-\mathrm{tr}(X^2) \geq 0$ for anti-Hermitian X. We name the fourth coordinate $\tau = x^4$ and use the spatial-first orientation $\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3 \wedge \mathrm{d}x^4$ on $\mathbb{R}^4$. It is the negative of the time-first orientation. The coordinate τ belongs to this Riemannian problem.

Physical input. In the Euclidean path-integral formulation of quantum gauge theory, stationary Euclidean fields contribute to quantum amplitudes with a leading semiclassical weight $\sim e^{-S_E/\hbar}$. This is cited physics from Tong, §2.3.1–2.3.3, with $\hbar$ restored. Fluctuations supply additional factors, so this weight is not a transition probability. Interpreting τ as Euclidean time uses this additional formulation of the quantum theory.

Lecture 15 gives $\star^2 = 1$ on Euclidean two-forms.

Definition 23.5 (Duality). A curvature is *self-dual* if $F = \star F$ and *anti-self-dual* if $F = -\star F$.

Its orthogonal parts $F_+ = (F + \star F)/2$ and $F_- = (F - \star F)/2$ obey $\star F_+ = F_+$ and $\star F_- = -F_-$. Either duality equation implies the vacuum field equation, since $D\star F = \pm DF = 0$ by Lecture 21's Bianchi identity. Real Minkowski two-forms have $\star^2 = -1$. Imposing either real duality equation there forces $F = 0$.

Reversing orientation changes the star's sign and also changes integration's sign. Their product leaves S_E unchanged. The action depends on the metric through the star. An integral of a fixed four-form instead changes sign with orientation.

23.3 Chern–Simons and the second Chern number

In Lecture 22, the local identity $F = \mathrm{d}A$ turned a curvature integral over two caps into an integral of their potential difference along the equator. For the second Chern form, we seek a local three-form whose derivative is $\operatorname{tr}(F \wedge F)$. Its boundary integral will relate the four-dimensional charge to an equatorial transition. For matrix-valued forms of degrees p and q, cyclicity becomes $\operatorname{tr}(B \wedge C) = (-1)^{pq} \operatorname{tr}(C \wedge B)$. This follows by interchanging the scalar differential forms in each matrix entry.

Definition 23.6 (Chern–Simons form)**.** In a local frame define the *Chern–Simons three-form* by

$$\mathrm{CS}(A) = \operatorname{tr}\left(A \wedge \mathrm{d}A + \frac{2}{3} A \wedge A \wedge A\right).$$

The coefficient $2/3$ compensates for the derivative of the cubic term. We denote the Maurer–Cartan form by ϑ_{MC}, distinct from a coframe θ^a.

Theorem 23.7 (The local transgression · *sketched; local identities proved*)**.** *The local forms satisfy*

$$\mathrm{d}\,\mathrm{CS}(A) = \operatorname{tr}(F \wedge F), \qquad \mathrm{CS}(\vartheta_{\mathrm{MC}}) = -\frac{1}{3} \operatorname{tr}(\vartheta_{\mathrm{MC}}^3), \qquad \vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h.$$

For a general frame change, the local gauge law is

$$\mathrm{CS}(A^h) = \mathrm{CS}(A) - \mathrm{d} \operatorname{tr}(\mathrm{d}h h^{-1} \wedge A) - \frac{1}{3} \operatorname{tr}(\vartheta_{\mathrm{MC}}^3).$$

The resulting global characteristic-class interpretation is cited.

Proof of the local identities. Write matrix wedge products as juxtaposition. The product rule gives $\mathrm{d} \operatorname{tr}(A\mathrm{d}A) = \operatorname{tr}((\mathrm{d}A)^2)$. For the cubic term it gives

$$\mathrm{d} \operatorname{tr}(A^3) = \operatorname{tr}(\mathrm{d}A\, A^2 - A\, \mathrm{d}A\, A + A^2 \mathrm{d}A) = 3 \operatorname{tr}(\mathrm{d}A\, A^2).$$

Moving the first A past the remaining total degree three changes sign. This proves the middle term's sign and also gives $\operatorname{tr}(A^4) = -\operatorname{tr}(A^4) = 0$. Consequently

$$\operatorname{tr}(F^2) = \operatorname{tr}((\mathrm{d}A)^2 + 2\mathrm{d}A\, A^2) = \mathrm{d}\,\mathrm{CS}(A).$$

The Maurer–Cartan equation is $\mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}}^2$. Substitution into the definition gives $(-1 + 2/3) \operatorname{tr}(\vartheta_{\mathrm{MC}}^3)$, which is the claimed pure-gauge value. More generally, a coefficient c in place of $2/3$ would give $3c$ in front of $\operatorname{tr}(\mathrm{d}A\, A^2)$. Matching the curvature square requires $3c = 2$, hence $c = 2/3$. ∎

For the general law set $B = h^{-1}Ah$. Expand $\mathrm{CS}(B + \vartheta_{\mathrm{MC}})$ using $\mathrm{d}B = h^{-1}\mathrm{d}A h - \vartheta_{\mathrm{MC}} B - B\vartheta_{\mathrm{MC}}$. Graded trace cyclicity groups the mixed terms into $-\mathrm{d} \operatorname{tr}(\vartheta_{\mathrm{MC}} \wedge B) = -\mathrm{d} \operatorname{tr}(\mathrm{d}h h^{-1} \wedge A)$. The unmixed terms give the two Chern–Simons forms above. This outlines the expansion. Nakahara, §11.5, supplies the general transgression framework. The local potential and its Chern–Simons form need not be global. On a closed connected oriented four-manifold, a nonzero integral of $\operatorname{tr}(F \wedge F)$ obstructs a global three-form primitive by Stokes. A local identity does not assert that such a primitive exists.

Definition 23.8 (Second Chern number)**.** For an SU(2) bundle over an oriented closed four-manifold, its *second Chern number* is

$$k = \int_M c_2(F) = \frac{1}{8\pi^2}\int_M \operatorname{tr}(F \wedge F).$$

Lecture 22's higher Chern forms show that this number is independent of the connection and metric. Its sign uses the orientation.

Definition 23.9 (Chern character)**.** The *Chern character* is the trace exponential, expanded in wedge powers:

$$\operatorname{ch}(F) = \operatorname{tr}\exp(\mathrm{i}F/(2\pi)).$$

The quadratic exponential term gives its degree-four component on an SU(2) bundle:

$$\operatorname{ch}_2(F) = -\frac{1}{8\pi^2}\operatorname{tr}(F \wedge F) = -c_2(F).$$

Thus our k is minus the integrated second Chern character. [3]

On S^4 the integer value of our Chern number can be read from an equatorial transition. Represent the two hemispheres by an inside and an outside four-ball as in the figure.

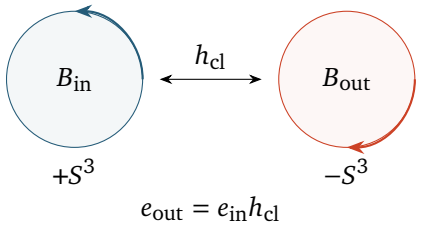


Figure 23.1: Two local Stokes integrals have opposite equatorial orientations. The circles represent the equator with signs relative to ∂B_{in} for the frame order $e_{\text{out}} = e_{\text{in}} h_{\text{cl}}$.

Theorem 23.10 (The clutching integer · *sketched; global construction cited*)**.** *Let* $h_{\text{cl}} \colon S^3 \to \mathrm{SU}(2)$ *glue the frames in the figure's order. Orient* SU(2) *as the outward unit sphere under* $(a_1, a_2, a_3, a_4) \mapsto a_4 I + \mathrm{i} a_a \sigma_a$*. Then* $k = -\deg h_{\text{cl}} \in \mathbb{Z}$*.*

Sketch. Use local frames over the two balls, and Stokes with their opposite boundaries. The gauge law and the vanishing integral of an equatorial exact form give

$$k = \frac{1}{8\pi^2}\oint_{S^3}\left(\operatorname{CS}(A_{\text{in}}) - \operatorname{CS}(A_{\text{out}})\right) = \frac{1}{24\pi^2}\oint_{S^3}\operatorname{tr}\left((h_{\text{cl}}^{-1}\mathrm{d}h_{\text{cl}})^3\right).$$

For the identity unit-quaternion map, at $a_4 = 1$ the Maurer–Cartan form is $\mathrm{i}\sigma_a \mathrm{d}a_a$. The Pauli products give cubic trace $12\mathrm{d}a_1 \wedge \mathrm{d}a_2 \wedge \mathrm{d}a_3$. The outward sphere volume there is its negative, since the outward normal is last in the ordered ambient coordinates.[4] Both forms are left invariant, so everywhere $\operatorname{tr}(\vartheta^3_{\text{MC}}) = -12\operatorname{vol}_{S^3}$. The sphere volume is $4\pi\int_0^\pi \sin^2\chi\,\mathrm{d}\chi = 2\pi^2$. Thus the normalized three-form $-\operatorname{tr}(\vartheta^3_{\text{MC}})/(24\pi^2)$ has integral one. Lecture 14's degree formula identifies the displayed integral as minus the degree. The global two-ball construction is the cited input from Nakahara, §10.5.5. ∎

[3] Nakahara's (11.54)–(11.55) and (12.90) use the integrated second Chern character as their instanton number.

[4] In Nakahara (10.124)–(10.125), the equatorial north-pole volume is $+\mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$. Our outward volume form at that pole is its negative.

On a region with boundary, the integral of the Chern form need not be an integer, just as the cap flux in Lecture 22 need not be integral.

The comparison with Lecture 22 records the local primitive and the boundary map in each dimension.

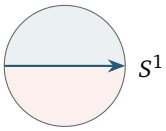
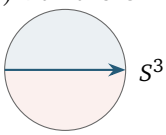
Line bundle on S^2 	SU(2) **bundle on S^4** 
$F = \mathrm{d}A$	$\operatorname{tr}(F \wedge F) = \mathrm{d}\,\mathrm{CS}(A)$
$e_S = e_N g_{NS}$	$e_{\text{out}} = e_{\text{in}} h_{\text{cl}}$
$g_{NS} \colon S^1 \to \mathrm{U}(1)$	$h_{\text{cl}} \colon S^3 \to \mathrm{SU}(2)$
$\dfrac{1}{2\pi \mathrm{i}} \oint g_{NS}^{-1} \mathrm{d} g_{NS}$	$\dfrac{1}{24\pi^2} \oint \operatorname{tr}((h_{\text{cl}}^{-1} \mathrm{d} h_{\text{cl}})^3)$
$C_1 = \operatorname{wind} g_{NS}$	$k = -\deg h_{\text{cl}}$

Figure 23.2: The two-cap method uses a local primitive in each frame and the stated equatorial frame transition. The line-bundle equator is the northern boundary, and the four-dimensional equator is ∂B_{in}.

For a field on $\mathbb{R}^4$, the same integer applies when a change of frame near infinity extends the connection smoothly over the compactification S^4. If its asymptotic potential is $h_\infty^{-1} \mathrm{d} h_\infty$, the outer frame uses the inverse transition, so $k = \deg h_\infty$. This requires the stated asymptotic and extension conditions. Finite action alone does not supply them. The BPST potential in Unit 23C exhibits this asymptotic form explicitly.

23.4 The action bound

For an adjoint-valued two-form B, write $\|B\|^2 = -\int_M \operatorname{tr}(B \wedge \star B)$. This is nonnegative and vanishes only for $B = 0$.

Theorem 23.11 (The instanton bound · *proved*)**.** *For an* SU(2) *connection over Euclidean* S^4*,*

$$S_E \geq \frac{8\pi^2}{g_{\text{YM}}^2} |k|.$$

Equality holds exactly for a self-dual or anti-self-dual curvature. In our negative-trace convention, a nonzero self-dual field has $k < 0$*, and a nonzero anti-self-dual field has* $k > 0$*.*

Proof. The Euclidean star is an isometry with square one on two-forms. Its two eigenspaces are orthogonal. Using the parts F_+ and F_- defined above gives

$$g_{\text{YM}}^2 S_E = \|F_+\|^2 + \|F_-\|^2, \qquad 8\pi^2 k = \|F_-\|^2 - \|F_+\|^2.$$

The sum of two nonnegative numbers is at least the absolute value of their difference. Equality holds exactly when one number vanishes. For $k > 0$ this requires $F_+ = 0$, while for $k < 0$ it requires $F_- = 0$. For $k = 0$ equality forces $F = 0$, which obeys both duality equations.

The same decomposition gives the square completion directly. Expand the square for either $s = +1$ or $s = -1$ to obtain

$$-\operatorname{tr}(F \wedge \star F) = -\frac{1}{2}\operatorname{tr}\big((F - s\star F) \wedge \star(F - s\star F)\big) - s\operatorname{tr}(F \wedge F).$$

After integration this gives $g_{\mathrm{YM}}^2 S_E = \frac{1}{2}\|F - s\star F\|^2 - s8\pi^2 k$. Choose s so that $-sk = |k|$. Nonnegativity proves the bound. Equality requires $F = s\star F$. Conversely, that equation removes the square and gives equality. For a nonzero dual field, $8\pi^2 k = -s\|F\|^2$, proving the stated signs. ∎

Definition 23.12 (Instanton). An *instanton* here is a smooth finite-action Euclidean connection with self-dual or anti-self-dual curvature.

On S^4, or under the stated extension from $\mathbb{R}^4$, the bound makes it an action minimum within its fixed integer sector.

The norm plane shows how a fixed charge constrains the two curvature components.

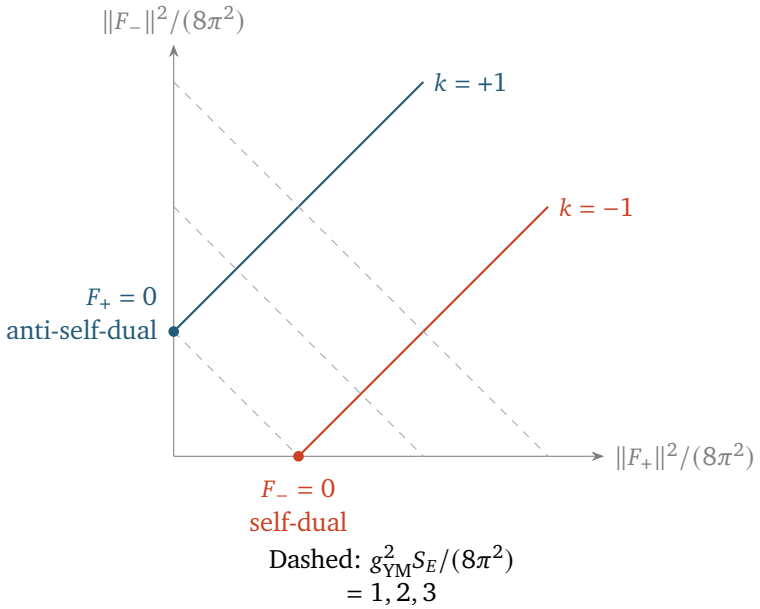


Figure 23.3: The coordinates are squared norms divided by $8\pi^2$, rather than spatial coordinates. Each fixed-charge line first meets an action contour on an axis, where one duality component vanishes.

The bound also compares the minimum actions in different integer sectors.

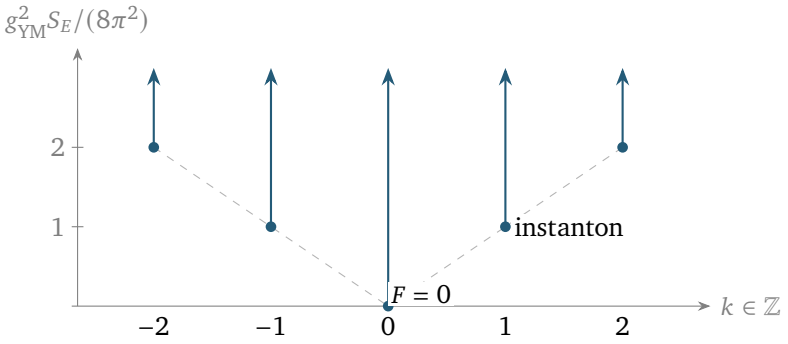


Figure 23.4: The bound is plotted at integer charges, with height $g_{\mathrm{YM}}^2 S_E/(8\pi^2)$. The vertical rays indicate the allowed inequality in each sector, and equality requires a dual curvature.

Self-check. State in one sentence what an instanton is and why it was introduced.

Answer. An instanton is a smooth finite-action Euclidean connection with self-dual or anti-self-dual curvature, introduced because the duality equation attains the least action in its fixed Chern-number sector on S^4, or under the stated smooth extension from $\mathbb{R}^4$. The decisive step is that one of $\|F_+\|^2$ and $\|F_-\|^2$ vanishes.

Unit 23C The BPST field, the topological term and principal connections

23.5 A charge-one field and a topological term

The *BPST instanton* is the following charge-one SU(2) example with size $\lambda > 0$. Recall $r^2 = \sum_{\mu=1}^{4} (x^\mu)^2$, with spatial indices $a, b, c \in \{1, 2, 3\}$. The potential is smooth at the origin and is given by

$$A = \frac{\mathrm{i}\sigma_a \left(x^4 \mathrm{d}x^a - x^a \mathrm{d}x^4 + \epsilon_{abc} x^b \mathrm{d}x^c\right)}{r^2 + \lambda^2}.$$

For $r > 0$, define the asymptotic unit-quaternion map

$$h_\infty : S^3 \to \mathrm{SU}(2), \qquad h_\infty(x/r) = \frac{x^4 I + \mathrm{i} x^a \sigma_a}{r}.$$

Its radial extension gives the same potential in the form

$$A = f(r)\vartheta_{\mathrm{MC}}, \qquad f(r) = \frac{r^2}{r^2 + \lambda^2}, \qquad \vartheta_{\mathrm{MC}} = h_\infty^{-1} \mathrm{d}h_\infty.$$

The scalar factor tends to one as $r \to \infty$. Thus the asymptotic pure gauge is explicit for this field.

The figure compares an exact exterior pure gauge in the plane with the BPST asymptotic gauge.

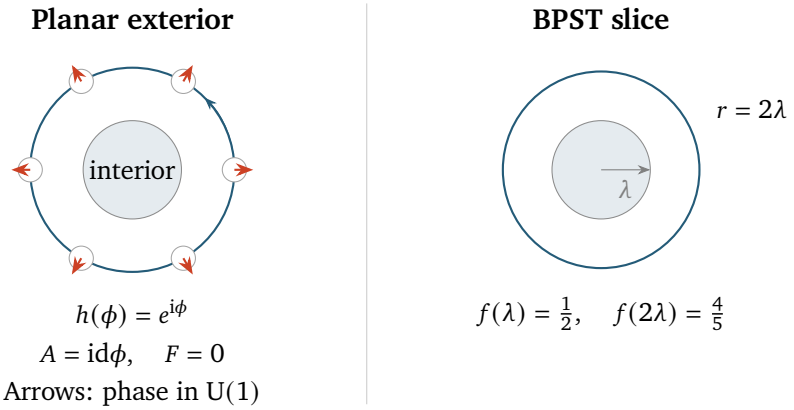


Figure 23.5: The planar exterior has $A = i\,d\phi$ and a single-valued phase of winding one. The BPST panel is a two-dimensional slice: its factor $f(r)$ approaches one, while its curvature remains nonzero at every finite radius.

Example 23.13 (The BPST curvature and action)**.** We compute the curvature, charge and action of this regular potential. We compute the component F_{14} from its derivative and commutator terms. We abbreviate the positive denominator by $d = r^2 + \lambda^2$ and write $t = x^4$. Here d is a scalar, distinct from the exterior derivative d. Let $\mathbf{x} = (x^1, x^2, x^3)$ and $\mathbf{b} = (t, x^3, -x^2)$. The potential has components

$$A_1 = \frac{i\boldsymbol{\sigma} \cdot \mathbf{b}}{d}, \qquad A_4 = -\frac{i\boldsymbol{\sigma} \cdot \mathbf{x}}{d}.$$

The Pauli commutator is $[i\boldsymbol{\sigma} \cdot \mathbf{u}, i\boldsymbol{\sigma} \cdot \mathbf{v}] = -2i\boldsymbol{\sigma} \cdot (\mathbf{u} \times \mathbf{v})$ for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$. Differentiating the components and applying this commutator gives

$$\partial_1 A_4 - \partial_4 A_1 = -\frac{2i\sigma_x}{d} + \frac{2i}{d^2}\boldsymbol{\sigma} \cdot (x^1\mathbf{x} + t\mathbf{b}),$$
$$[A_1, A_4] = \frac{2i}{d^2}\boldsymbol{\sigma} \cdot (\mathbf{b} \times \mathbf{x}).$$

The vector terms cancel because

$$x^1\mathbf{x} + t\mathbf{b} + \mathbf{b} \times \mathbf{x} = r^2(1, 0, 0).$$

Hence the coefficient of $dx^1 \wedge dx^4$ is

$$F_{14} = -\frac{2i\sigma_x}{d} + \frac{2ir^2\sigma_x}{d^2} = -\frac{2i\lambda^2\sigma_x}{d^2}.$$

The spatial pair gives $F_{23} = 2i\lambda^2\sigma_x/d^2$. Cyclic permutation of the spatial labels gives the other pairs. The full curvature is

$$F = \frac{2i\lambda^2\sigma_a}{(r^2 + \lambda^2)^2}\left(\frac{1}{2}\epsilon_{abc}\,dx^b \wedge dx^c - dx^a \wedge dx^4\right).$$

For example, the coefficient pair multiplying σ_x is $dx^2 \wedge dx^3 - dx^1 \wedge dx^4$. The spatial-first star interchanges these two basis forms, so their difference is anti-self-dual. The other cyclic pairs have the same property. Taking the fundamental trace gives

$$\operatorname{tr}(F \wedge F) = -\operatorname{tr}(F \wedge \star F) = \frac{48\lambda^4}{(r^2 + \lambda^2)^4}\,\mathrm{vol}_{\mathbb{R}^4}.$$

Radial integration, followed by $w = r^2/\lambda^2$, yields

$$\int_{\mathbb{R}^4} \operatorname{tr}(F \wedge F) = 96\pi^2\lambda^4 \int_0^\infty \frac{r^3 \mathrm{d}r}{(r^2+\lambda^2)^4} = 48\pi^2 \int_0^\infty \frac{w \mathrm{d}w}{(1+w)^4} = 8\pi^2.$$

The last integral is $1/6$, by writing $w = (1+w) - 1$ and integrating the two powers. Consequently $k = 1$ and $S_E = 8\pi^2/g_{\mathrm{YM}}^2$, independent of the size.

The asymptotic map h_∞ has degree +1 in the stated orientation. At infinity the pure-gauge Chern–Simons integral therefore gives $k = -\int_{S^3} \operatorname{tr}((h_\infty^{-1}\mathrm{d}h_\infty)^3)/(24\pi^2) = +1$. The clutching map has the inverse order. The change of frame $h_{\mathrm{cl}} = h_\infty^{-1}$ makes the outer asymptotic potential zero. Its degree is -1, so the clutching formula gives the same charge. Reversing to time-first orientation makes this same field self-dual and changes k to -1.[5]

The normalized radial contribution for this example is plotted in the figure. Its maximum occurs at $u = \sqrt{3/5}$, so the size λ fixes the radial scale of the density.

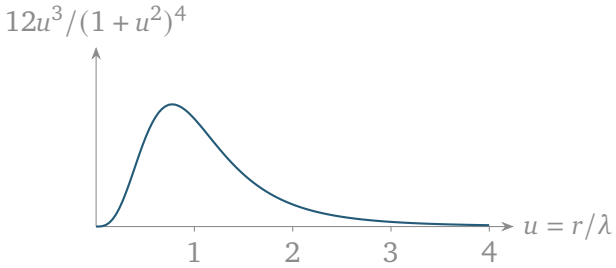


Figure 23.6: For $u = r/\lambda$, the BPST Chern-number contribution per du is $12u^3/(1+u^2)^4$. Its integral over $[0,\infty)$ is 1, and the plot shows the initial interval $0 \le u \le 4$.

Proposition 23.14 (BPST normalization · *sketched*). *For size $\lambda > 0$ on spatial-first $\mathbb{R}^4$, the regular BPST field is anti-self-dual, extends over the conformal compactification S^4, and has $k = +1$ and $S_E = 8\pi^2/g_{\mathrm{YM}}^2$. Reflection of one coordinate gives its self-dual partner with $k = -1$ and the same action.*

Extension and orientation

The extension over the compactification is a cited part of the BPST construction in Nakahara, §10.5.5, and Tong, §2.3.1. We assume this asymptotic regularity when treating an $\mathbb{R}^4$ field as a connection on S^4. Finite action alone is not used as a proof of that extension. The Euclidean star on two-forms is unchanged by a conformal metric rescaling, because its scale factor in dimension four is the zeroth power. Thus stereographic compactification preserves the duality equation and action. Give $S^4 \subset \mathbb{R}^5$ its outward orientation. Let $e_1, \dots, e_5$ be the standard basis of $\mathbb{R}^5$ and put $r^2 = \sum_{\mu=1}^4 (x^\mu)^2$. Use the inverse south-pole stereographic chart $X\colon \mathbb{R}^4 \to S^4 \setminus \{-e_5\}$, given by

$$X(x) = \frac{(2x^1, 2x^2, 2x^3, 2x^4, 1-r^2)}{1+r^2}.$$

At $x = 0$, the outward normal is e_5 and the four coordinate tangent vectors are $2e_i$. The ordered determinant is +16. The inverse $x^i = X^i/(1+X^5)$, for $i \in \{1,2,3,4\}$, shows that this

[5] Lecture 4 wrote $U = aI - \mathrm{i}v^a\sigma_a$. Our oriented quaternion coordinates there are $(a_4, \mathbf{a}) = (a, -\mathbf{v})$. Tong calls his time-first BPST field self-dual with $\nu = +1$. Our spatial-first orientation and translated anti-Hermitian field give anti-self-duality and $k = +1$.

is a smooth chart. Since $\mathbb{R}^4$ is connected, it preserves the spatial-first orientation everywhere. Consequently the computed charge on $\mathbb{R}^4$ is also the charge on this outward S^4.

Definition 23.15 (Theta term). For a fixed real coefficient θ_{YM}, distinct from the Maurer–Cartan form ϑ_{MC}, the *theta term* is

$$S_\theta = \frac{\theta_{\mathrm{YM}}}{8\pi^2}\int_M \operatorname{tr}(F\wedge F) = \theta_{\mathrm{YM}}k.$$

Locally its density is $\theta_{\mathrm{YM}}\mathrm{d}\,\mathrm{CS}(A)/(8\pi^2)$. Its variation is $\delta\operatorname{tr}(F\wedge F) = 2\mathrm{d}\operatorname{tr}(a\wedge F)$, using $DF = 0$. It therefore leaves the field equation unchanged for compactly supported variations, or on a closed manifold. On a manifold with boundary, boundary conditions still matter.

Physical input. In the quantum path integral, an added real-time action S_θ multiplies a field's weight by $e^{\mathrm{i}S_\theta/\hbar}$, as in Tong, §2.2, with $\hbar$ restored.

For integer k, this factor is $e^{\mathrm{i}(\theta_{\mathrm{YM}}/\hbar)k}$. It is unchanged when θ_{YM} increases by $2\pi\hbar$. Thus the dimensionless angle is $\theta_{\mathrm{YM}}/\hbar$. The topological term can change the relative quantum weights of sectors even though its bulk variation vanishes.

Self-check. For the BPST potential, take $\lambda = 2$ and $(x^1, x^2, x^3, x^4) = (1, 2, -1, 3)$. Compute $F_{23} = \partial_2 A_3 - \partial_3 A_2 + [A_2, A_3]$ by differentiating the potential before evaluating it at this point.

Answer. The denominator is $d = 19$. Differentiating first retains the derivatives of d and gives

$$\partial_2 A_3 - \partial_3 A_2 = \frac{\mathrm{i}}{361}(28\sigma_x - 2\sigma_y - 14\sigma_z),$$
$$[A_2, A_3] = \frac{\mathrm{i}}{361}(-20\sigma_x + 2\sigma_y + 14\sigma_z).$$

The last two Pauli components cancel, leaving $F_{23} = 8\mathrm{i}\sigma_x/361$. This agrees with the coefficient in the full curvature.

23.6 A connection on the principal bundle

The local potentials also have an intrinsic description on a principal bundle. Lecture 20's principal bundle $P \to M$ carries a right action of a Lie group G. Its Lie algebra is $\mathfrak{g}$.

Definition 23.16 (Fundamental vertical vector). For $\xi \in \mathfrak{g}$, the *fundamental vertical vector* at $p \in P$ is $\xi_p^{\#} = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_0 p\exp(t\xi)$.

It is tangent to the fibre.

Definition 23.17 (Principal connection). A *principal connection* is a smooth $\mathfrak{g}$-valued one-form ω_P on P with

$$\omega_P(\xi^{\#}) = \xi, \qquad R_a^*\omega_P = \mathrm{Ad}_{a^{-1}}\,\omega_P \quad (a \in G),$$

where $R_a(p) = pa$ and $\mathrm{Ad}_a\,\xi = a\xi a^{-1}$ for a matrix group.

Definition 23.18 (Gauge transformation of a principal bundle). Gauge transformations are bundle automorphisms over the identity that commute with the right action.

Lecture 21 represents them by compatible maps $u_j = g_{ij}^{-1}u_i g_{ij}$. A global map $M \to G$ suffices in a chosen trivialization.

Remark 23.19 (Principal connections)**.** The kernel of ω_P gives the horizontal tangent spaces, complementary to the vertical tangent spaces. In a local section $s_i : U_i \to P$, the potential is $A_i = s_i^* \omega_P$. For $s_j = s_i g_{ij}$ these potentials obey Lecture 21's frame law. A representation $\rho : G \to \mathrm{GL}(V)$ induces on the associated bundle the local derivative $\mathrm{d} + \rho_*(A_i)$, where ρ_* is the representation's differential at the identity. Thus one principal connection supplies the potentials in all representations. These intrinsic-to-local facts are cited from Tu, §28.1, §29.4 and §31.3–31.5.

Summary. The fundamental anti-Hermitian trace fixes the Minkowski prefactor and gives $D\star F = 0$. For U(1) the different trace normalization recovers Maxwell's action. Locally $\mathrm{d}\,\mathrm{CS}(A) = \mathrm{tr}(F \wedge F)$. The Chern–Simons form need not be global. On S^4, $k = -\deg h_{\mathrm{cl}}$ in the stated two-frame order. The Euclidean bound is $S_E \geq 8\pi^2|k|/g_{\mathrm{YM}}^2$, saturated exactly by dual fields. Regular spatial-first BPST is anti-self-dual with $k = +1$. The action needs the metric and is orientation independent. The Chern number has the opposite dependence. Lecture 24 returns to line-bundle connections generated by quantum eigenstates.

Reading. Tu, *Differential Geometry*, §28.1, §29.4 and §31.3–31.5. Nakahara, §10.5.4–10.5.5, §11.3.1 (11.54)–(11.55), §11.5 and §12.6 (12.90). Tong, *Gauge Theory*, §2.1–2.3, for the action, topological phase and semiclassical Euclidean input.